

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Quality Engineering and test infrastructure professional with over **6+ years** of experience building scalable integration, end-to-end, API, and CI/CD testing systems that improve engineering confidence and production readiness.
- Strong software engineering experience with **Python, TypeScript, JavaScript, C#, Java, React-oriented frontend testing, REST APIs, and SQL**, emphasizing clean, maintainable code and reusable automation architecture.
- Hands-on expertise with **Playwright, Cypress, Pytest, Jest, Selenium WebDriver, RESTAssured, and Cucumber**, developing fast and reliable frontend/backend test frameworks that can be adopted across multiple engineering teams.
- Experienced building and optimizing test execution in **CI/CD pipelines**, shortening feedback loops, improving test parallelization, enforcing automated quality guardrails, and maintaining stable integration and E2E environments.
- Strong background in **data mocking, service isolation, reusable fixtures, test-data utilities, microservice validation, and coverage-gap analysis**, enabling high-fidelity testing of complex application flows at scale.
- Experienced applying **Claude, Cursor, Copilot, and AI-assisted quality workflows** for intelligent test generation, regression analysis, flaky-test diagnosis, debugging, code maintenance, and faster feature delivery.
- Ownership-driven engineer experienced troubleshooting complex failures, performing root-cause analysis, monitoring test-suite and CI health, partnering with feature teams, and implementing systemic improvements that prevent future regressions.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Full Stack:** Python, Django, Go, TypeScript, JavaScript, React, C#, Java, SQL, REST APIs, Microservices, Object-Oriented Programming, Clean Code.
- **Test Frameworks & Tooling:** Playwright, Cypress, Pytest, Jest, Selenium WebDriver, RESTAssured, Cucumber BDD, NUnit, XUnit, JUnit, TestNG, Integration/E2E Framework Design.
- **Test Infrastructure:** Data Mocking, Service Isolation, Fixtures, Test Data Management, Reusable Libraries, API Clients, Environment Abstractions, Coverage Gap Analysis, Repository Test Standards.
- **CI/CD & Infrastructure:** Buildkite, GitHub Actions, Jenkins, Azure DevOps, GitLab CI, Git, AWS, Kubernetes, Terraform, Helm, Docker, Parallel Test Execution, Automated Guardrails.
- **Data & Distributed Systems:** MySQL, PostgreSQL, Redis, Kafka, SQL Validation, Event-Driven Systems, Backend Services, Data Consistency, Message Processing, Integration Testing.
- **AI-Driven Quality:** Claude, Cursor, GitHub Copilot, AI Test Generation, Automated Regression Detection, Flaky Test Analysis, Intelligent Debugging, AI-Assisted Code Refactoring, Coverage Optimization.
- **Engineering Practices:** Root Cause Analysis, Test Suite Health, CI Monitoring, Code Reviews, Repository Standards, Unit Testing, Production Readiness, Agile/Scrum, Operational Excellence, Continuous Improvement.

Test Automation Engineer

PCMI LLC (Full Time)

Oct 2024 – Present

Boston, MA, USA

- Own and enhance integration and end-to-end automation frameworks using **Python/C#, Playwright, Pytest-style utilities, REST APIs, and SQL**, building reusable infrastructure that supports reliable validation across frontend and backend application stacks.

- Develop shared testing utilities including fixtures, API clients, data builders, service abstractions, reusable assertions, and mocking capabilities that simplify complex scenarios and make automated testing easier for engineering teams to adopt.
- Integrate automated suites with **Azure DevOps, TeamCity, Git, Docker, Kubernetes, and CI/CD pipelines**, optimizing parallel execution and feedback loops while enforcing automated quality gates before changes progress toward production.
- Investigate flaky tests and CI failures using logs, browser traces, API responses, SQL diagnostics, and environment telemetry, performing root-cause analysis and implementing systemic fixes instead of masking instability with retries.
- Identify and backfill critical coverage gaps across complex workflows using service isolation, controlled test data, API-level validation, and targeted end-to-end scenarios to increase confidence without unnecessarily expanding suite runtime.
- Leverage **Claude, Cursor, and Copilot** to accelerate test generation, analyze regression failures, refactor automation code, identify flaky patterns, and improve framework maintainability while independently reviewing AI-generated changes.
- Partner with feature and DevOps teams on test architecture, pull-request reviews, repository conventions, CI health, and production-readiness improvements while monitoring suite reliability, execution duration, failures, and recurring regression patterns.

Environments: Python, C#, Playwright, Pytest, REST APIs, SQL Server, Git, Azure DevOps, TeamCity, Docker, Kubernetes, Terraform, CI/CD, Data Mocking, Service Isolation, Fixtures, Claude, Cursor, Copilot, Flaky Test Analysis, Root Cause Analysis, Agile/Scrum.

SDET / QA Automation Engineer

Dec 2023 – Sept 2024

Client: CVS

Boston, MA, USA

- Built scalable frontend, API, and integration automation using **TypeScript, JavaScript, Playwright, Cypress, RESTAssured, and SQL**, validating complex React-style client workflows and distributed backend services.
- Developed reusable test framework components, fixtures, API abstractions, test-data utilities, and service mocks to isolate dependencies, increase scenario fidelity, and improve maintainability across frequently executed regression suites.
- Integrated automation with **Jenkins, GitLab/GitHub, Git, and CI/CD pipelines**, enabling parallel execution, rapid developer feedback, automated regression checks, and consistent quality signals for high-frequency software delivery.
- Troubleshoot intermittent tests and pipeline failures using Splunk, browser/network traces, API logs, database queries, and execution artifacts, converting root-cause findings into framework improvements that prevented recurring regressions.
- Validated backend workflows across REST services, asynchronous processing, databases, and integration boundaries while using controlled test data and dependency isolation to distinguish product defects from environment or test instability.
- Used AI-assisted tools such as **Copilot, Claude, and Cursor-style workflows** to generate test scaffolding, improve coverage, diagnose failed automation, and refactor reusable components while maintaining engineering review standards.
- Worked closely with software engineers and platform teams to identify testing gaps, strengthen repository conventions, review automation changes, and improve the health, performance, and reliability of CI and regression infrastructure.

Environments: TypeScript, JavaScript, Playwright, Cypress, Selenium WebDriver, RESTAssured, Postman, SQL, Jenkins, GitLab/GitHub, Git, CI/CD, Splunk, Data Mocking, Service Virtualization, AI-Assisted Testing, Flaky Test Analysis, Agile/Scrum.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Developed integration and end-to-end automation using **Python, Java, Selenium WebDriver, Cucumber BDD, Pytest/JUnit-style frameworks, REST APIs, and SQL** across enterprise frontend and backend applications.
- Built modular framework components, reusable fixtures, service clients, test-data utilities, and shared libraries while applying clean-code principles to make automation easier to maintain and extend across multiple application areas.

- Validated complex backend workflows involving APIs, databases, asynchronous processing, and event-driven integrations while using controlled dependencies and representative test data to improve coverage of high-risk application flows.
- Integrated automated suites with **Jenkins, Git, Maven, Docker, and CI/CD pipelines**, supporting continuous execution, automated build validation, regression feedback, and repeatable test environments across software releases.
- Investigated flaky automation and integration failures through application logs, API traces, SQL queries, timing analysis, and debugging, then refactored test or framework components to eliminate recurring failure patterns.
- Contributed to repository and testing standards by reviewing automation code, improving framework organization, identifying coverage gaps, and reducing duplicated or low-value tests that increased maintenance without meaningful risk protection.
- Collaborated with software and platform engineers to evaluate tooling, prototype improvements, strengthen test infrastructure, and apply root-cause findings to broader reliability and operational-quality improvements.

Environments: Python, Java, Selenium WebDriver, Cucumber BDD, Pytest/JUnit, REST APIs, SQL, Jenkins, Git, Maven, Docker, CI/CD, Service Isolation, Test Data Management, Framework Architecture, Flaky Test Analysis, Root Cause Analysis, Agile/Scrum.

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed maintainable functional, integration, API, and end-to-end automation using **Java, JavaScript, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL** across enterprise applications.
- Created reusable page objects, common test utilities, test-data builders, service abstractions, and shared validation components to reduce duplication and provide consistent automation patterns across regression suites.
- Performed backend and integration testing across APIs, databases, and dependent services using Postman, REST/SOAP interfaces, JSON/XML, and SQL while validating data consistency and business-critical application flows.
- Integrated automated tests with **Jenkins, Git, and CI/CD pipelines**, enabling repeatable build verification, scheduled regression execution, and earlier identification of software failures during iterative development.
- Diagnosed failed and intermittent tests using logs, screenshots, API responses, database records, and debugging techniques while working with developers to isolate root causes and implement lasting corrective changes.
- Improved regression efficiency by identifying redundant scenarios, strengthening high-value coverage, organizing test suites for maintainability, and refactoring brittle components that generated unnecessary CI noise.
- Collaborated with development and QA teams during code reviews, Agile ceremonies, and release activities to improve automation standards, knowledge sharing, test reliability, and shared ownership of production-quality software.

Environments: Java, JavaScript, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, REST APIs, SQL, Jenkins, Git, CI/CD, Reusable Test Utilities, Integration/E2E Testing, Test Suite Maintenance, Root Cause Analysis, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.